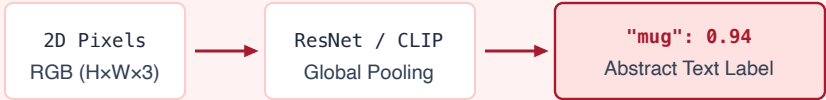


SPATIAL AFFORDANCE TOKENIZATION VS. PASSIVE DIGITAL CLASSIFICATION

Transforming 2D camera pixels into metric SE(3) coordinates, contact normals, and physical affordances

PASSIVE DIGITAL CLASSIFICATION (DISEMBODIED ML)



- ❌ **Zero Metric Scale:** Cannot deduce distance in meters or gripper aperture.
- ❌ **Zero Kinematic Utility:** Motor controller cannot actuate on a semantic string.
- ❌ **Blind to Geometry:** Ignores surface normals, collision hulls, and center of mass.

SPATIAL AFFORDANCE TOKENIZATION (PHYSICAL AI)



- ✓ **Metric Grounding:** Explicit coordinate frames relative to base frame (T_{base_cam}).
- ✓ **Direct Kinematic Authority:** Feeds inverse kinematics and trajectory decoders.
- ✓ **Mathematical Safety Barrier:** Bounding hulls supply distance d_{clear} to MCU CBF.

THE 4 ESSENTIAL PHYSICAL AFFORDANCE TOKENS EMITTED BY EMBODIED PERCEPTION

1. METRIC GRASP POSE (SE(3))

$$T_{grasp} = [R \mid p]$$

- Translation $p = [x, y, z]^T$ (m)
- Orientation $q = [qw, qx, qy, qz]$
- Gripper width aperture w_{grip}

$$\begin{aligned} x &= +0.342 \text{ m} \pm 0.004 \\ y &= -0.118 \text{ m} \pm 0.003 \\ z &= +0.085 \text{ m} \pm 0.002 \end{aligned}$$

2. CONTACT SURFACE GEOMETRY

$$\hat{n} = [n_x, n_y, n_z]^T, \kappa$$

- Unit surface normal vector $\hat{n}$
- Local principal curvature κ
- Approach alignment axis v_{app}

$$\begin{aligned} \hat{n} &= [0.02, 0.98, -0.19] \\ \text{Approach: } \angle(v, \hat{n}) &< 5^\circ \\ \text{Contact patch: Flat} \end{aligned}$$

3. MATERIAL & FRICTION TENSOR

$$\mu_{est}, K_{stiff}, m_{est}$$

- Friction coefficient $\hat{\mu} \approx 0.45$
- Material compliance K_{stiff}
- Estimated mass $m_{payload}$

$$\begin{aligned} \mu &= 0.52 \text{ (Ceramic)} \\ F_{normal_min} &= 4.2 \text{ N} \\ \text{Payload: } &0.380 \text{ kg} \end{aligned}$$

4. OBSTACLE BARRIER ENVELOPE

$$d_{clear}(x) \geq d_{safe}$$

- Dynamic bounding box B_{obs}
- Velocity vector v_{obs}
- Direct feed to MCU CBF

$$\begin{aligned} d_{clear} &= 0.142 \text{ m} \\ d_{stop} &= 0.088 \text{ m} \\ \text{Safety Margin: } &+54 \text{ mm} \end{aligned}$$